

ECON 101: Macroeconomics

Solutions – Quiz 5

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1. Correct answer: (C)

The Bank of Canada's main monetary policy tool is the **target overnight interest rate**. This is the interest rate it influences directly in order to affect other market interest rates, borrowing, spending, and inflation.

2. Correct answer: (C)

A decrease in the target overnight rate lowers short-term interest rates in the economy. Lower interest rates reduce borrowing costs for households and firms, which stimulates consumption and investment spending.

3. Correct answer: (B)

The monetary transmission mechanism works partly through interest rates. When interest rates change, firms adjust investment and households adjust spending, which then affects aggregate demand.

4. Correct answer: (B)

If Canadian interest rates rise, Canadian financial assets become more attractive to foreign investors. Demand for the Canadian dollar rises, so the dollar tends to **appreciate**.

5. Correct answer: (C)

Contractionary monetary policy raises interest rates, which reduces consumption, investment, and net exports. This lowers aggregate demand, so the AD curve shifts to the **left**.

Given:

- Potential GDP: $Y^* = 2,500$
- Initial actual GDP: $Y = 2,700$
- Investment falls by \$100 billion
- $MPC = 0.75$

(a) Calculate the spending multiplier (3 marks)

The simple spending multiplier is:

$$k = \frac{1}{1 - \text{MPC}}$$

Substitute the given MPC:

$$k = \frac{1}{1 - 0.75}$$

$$k = \frac{1}{0.25} = 4$$

$$\boxed{k = 4}$$

Answer: The spending multiplier is $\boxed{4}$.

(b) Calculate the total change in equilibrium GDP (4 marks)

The formula for the total change in equilibrium GDP is:

$$\Delta Y = k \times \Delta I$$

We know:

$$k = 4, \quad \Delta I = -100$$

Substitute:

$$\Delta Y = 4 \times (-100)$$

$$\Delta Y = -400$$

$$\boxed{\Delta Y = -400}$$

Answer: Equilibrium GDP decreases by $\boxed{\$400 \text{ billion}}$.

(c) Calculate the new level of equilibrium GDP (2 marks)

Initial GDP:

$$Y_0 = 2,700$$

Change in GDP:

$$\Delta Y = -400$$

New equilibrium GDP:

$$Y_1 = Y_0 + \Delta Y$$

$$Y_1 = 2,700 - 400 = 2,300$$

$$Y_1 = 2,300$$

Answer: The new equilibrium GDP is $2,300$.

(d) AD–AS diagram explanation (3 marks)

Students should draw an AD–AS diagram with:

- a downward-sloping AD_0 curve,
- an upward-sloping SRAS curve,
- a vertical LRAS curve at $Y^* = 2,500$,
- an initial equilibrium where AD_0 intersects SRAS at $Y = 2,700$,
- a leftward shift of AD from AD_0 to AD_1 ,
- a new short-run equilibrium at $Y = 2,300$.

Interpretation:

- Initially, the economy is producing above potential GDP, so there is an **inflationary gap**.
- The increase in the policy rate reduces investment, which shifts AD left.
- In the short run, both output and the price level fall.

Required labels:

- AD_0 , AD_1
- SRAS
- LRAS
- Initial equilibrium output 2,700
- New equilibrium output 2,300

(e) Explain how this monetary policy reduces inflation in the medium run (3 marks)

Step-by-step:

1. The Bank of Canada raises the policy interest rate.
2. Higher interest rates reduce borrowing and lower investment spending.
3. Lower investment reduces aggregate demand.
4. The AD curve shifts left.
5. Output falls from 2,700 to 2,300.
6. Because the economy was initially above potential GDP ($2,700 > 2,500$), there was upward pressure on prices.
7. Lower aggregate demand reduces this inflationary pressure.
8. In the medium run, slower demand growth helps inflation move back toward the Bank of Canada's target.

Important point: This policy helps reduce inflation because it closes the inflationary gap and weakens excess demand in the economy.

Contractionary monetary policy reduces inflation by lowering aggregate demand.
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End of Solution Key